Symmetric PKM: independent certificates and retained work

2026-11-05 · G1–G6 continuation · software prerequisite report

Scientific acceptance is incomplete. G1–G3/G6 partial; G4/G5 unrun; zero scientific panels; comparison admission false. There is no admitted complete-cost, GAE, scaling or global-novelty result. Original acceptance remains at PKM_GOALS.md; no threshold is weakened.

PR2949 merged; unpublished continuation preserved

PR2949 merged on November 1 as 8b147fd36e71990d9f44eb2cbfa435aa769cefc5. Maintenance stopped. All unpublished local commits survive on codex/pkm-certificates-accounting-20261105, without rebase or force-push. Original inputs, frozen evidence, receipt directories, reports and both histories of append-only logs remain preserved. This continuation adds the independent scalar replay/five supporting Lean contracts and then the G2 emitter corrections.

The separate native scalar reference uses rightmost pivots, Euclidean inversion and original-row provenance. It checks useful rank/RREF and directly verifies original-row recovery weights or a target-normalized kernel witness. Kernel ambiguity concerns the retained scalar equations, not multiple curve logarithms. Shared setup/encoder and Rust semantics remain outside the formal proof.

Collected repaired-source native run

Existing supervisor49719 ran source b022566e0c6723fef4695a59e054ae4fbdcd1fd0; it was collected without restart. Its 23examples, strict lint/format, ordinary CLI build, actual checkpoint-refusal message/expected exit1, backlog checks and library build pass. The receipt verifies17 source/fixture objects and6 actual binaries.

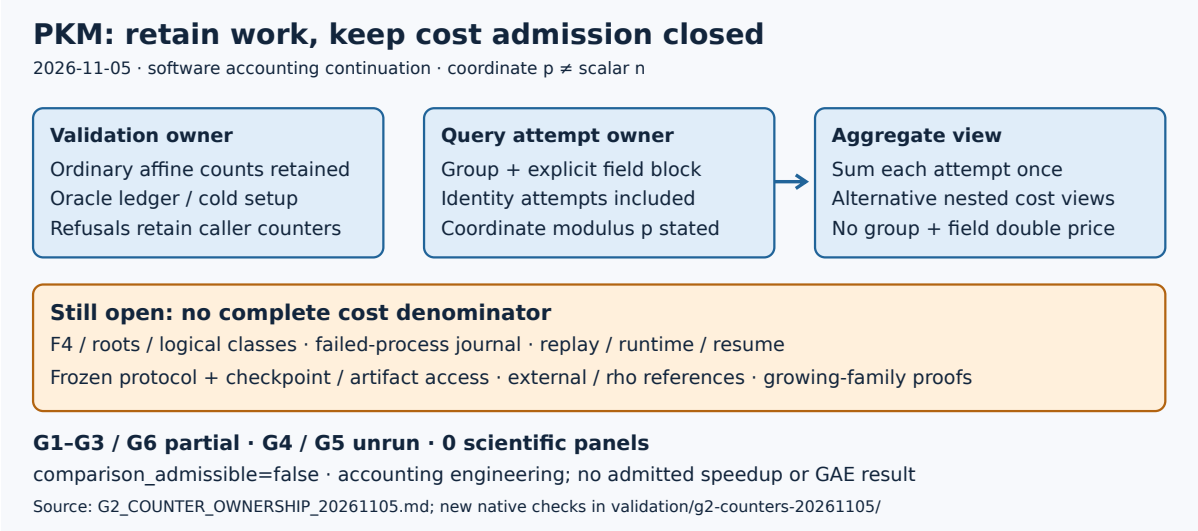
The library declares5848tests. Execution ran2026-10-30T20:48:46Z–21:28:46Z after compilation, then capped at2400seconds. Observed2272passes,1failure,48ignored,1JVcase active and3526not reached; these reconcile to5848. Timeout wrapper/supervisor124 is not a known Cargo direct exit. There is no final suite summary or full-suite pass. The parser includes3 should-panic annotations.

The failed chain-SAT test reproduces with the actual generated binary: atline1105, Some190568 disagrees with expected None. The implementation supports up to15summands while that assertion expects10unsupported. The source is identical to maincc5addeba; no unrelated assertion was changed, skipped or waived. The delimiter compilation defect is repaired; the separate assertion and unchangedJV runtime limit remain named blockers. Runtime cause is unknown; no deadlock or mathematical obstruction is inferred.

Raw library log221629bytes, SHA256 41ca32cee963b13adb6b5638abc52191de20f676b038874957a6b06e5e5ebebf, is **NOT_UPLOADED**. Exact local path, bounded excerpt and terminal records: validation/repaired-main-cc5addeba/final-20261105/receipt.json. Prior different-hash caps and failed attempts remain unchanged. Logs over64KiB are not split, compressed or given fabricated storage locations.

Formal scope and exact-object reuse

The final18-declaration/zero-trusted audit and official module kernel replay against imports remain valid for exactly unchanged proof/gate/dependency objects; no new proof run or –fresh replay is claimed. Lean4.34.1, Mathlib d13f23b723b8a846827a245b89c10fc7d3f11612. Only propext, Classical.choice and Quot.sound. The five contracts construct scalar coordinates for an exact generated cyclic group/prime-cardinality group and prove row recovery/kernel implications. Rust representation, polynomial correspondence and quantified complete solving/extraction-cost bridges remain open. See proof-map.md and integration-0cc5/proof-receipt.json.



New G2 implementation and verification

Source c41dc4fb2 retains validation reference counts, emits explicit per-attempt/aggregate query field and reduction blocks, and retains identity-query work. Shared GroupOps serialization remains unchanged. Aggregate totals are views of the same attempt work; group calls and their nested field calls are alternative pricing views. Helper/reference/scalar inverse diagnostics already have expanded multiplication children; ordinary-query inverse internals remain uncounted. Coordinate p=17 and scalar n=19 are separate units in the development fixture.

Source-pinned run1781 is terminal0. All18oracle tests pass (including3new counter controls), strict clippy/ rustfmt pass, and the rebuilt CLI refuses the controlled nonexistent input with the actual checkpoint-refusal message and expected exit1. The17 committed objects are clean/equal; new binary/capture hashes and terminals are in validation/g2-counters-20261105/receipt.json. Unchanged library, legacy-example, backlog and formal objects reuse exact-object evidence, including the library failure/cap. These are software checks, not a scientific panel. No native job remains running.

This source intentionally changes producer telemetry/setup; the earlier identical-producer-prefix assertion applies to faade, not this revision. Request/output schemas remain v1 with additive fields; replay v2 checks equations/certificates, not the completeness of counters. Canonical off-curve refusals retain caller counters, but top-level failed calls still lack a durable journal. FastCurve setup, F4/root/logical inventory, whole-process/replay costs, qualified runtime/resume and external/rho qualification remain incomplete. No price or end-to-end total is computed.

All six requirements retained

The complete acceptance criteria remain unchanged in PKM_GOALS.md. G1 needs the frozen complete-output corpus; G2 complete work coverage/pricing and durable failures; G3 equal-domain/external comparisons; G4 a held-out measured component gain under the original cost/memory thresholds; G5 predeclared depth and field-size panels; G6 qualified cold single-target comparisons and the missing central proof bridges. G1–G3/G6 stay partial, G4/G5 unrun. Software merge, toy controls and model arithmetic do not complete them.

No approved checkpoint/artifact/lab identity or qualified scientific execution/resume route is configured. No panel, funding, lease, IAM, quota or pool change occurred. The full-panel card remains blocked/unfunded/ unimplemented at its historical pin; checkpoint/protocol/runtime gates remain unchanged.

Sources, evidence and visual scope

Primary PKM2016 Algorithm1/§3.2: people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf; symmetry, dependence and LA are prior work. Sources checked before the counter fix include the repository measurement-unit contract, actual GroupOps/PrimeCurve code, G2_ACCOUNTING_AUDIT.md and G2_RECHECK_A030.md. This is accounting/correctness engineering, not a new relation algorithm or priority claim. G2_COUNTER_OWNERSHIP_20261105.md details ownership and known missing classes.

This diagram explains ownership because no comparable admitted measurements exist. No scientific numeric value, new curve or isogeny edge was admitted; canonical IC scoreboard/timeline, curves and performance graphs remain unchanged. Old reports/PDFs are preserved. Exact-head gate,size,artifacts, applicable reviews/resolved threads and a SHA guard remain necessary for any new PR merge; pending/ absent checks are not passes.